Chapter 2

Vapor Pressure – Inorganic Compounds

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TABULATION RESULTS

The tabulation is arranged by alphabetical order (Ac, Al, Am, ……) to provide ease of use in quickly locating the data by using the chemical formula. The compound name, chemical abstracts registry number (CAS No), and Antoine coefficients are provided in the adjacent columns. The range of application is denoted by minimum and maximum temperatures (TMIN and TMAX). Temperatures outside the range of application should not be used. The next column provides the code for the tabulation, which is based on both experimental data and estimated values.

In preparing the tabulation, a literature search was conducted to identify data source publications (53–115). Both experimental values for the property under consideration and parameter values for estimation of the property are included in the source publications. The publications were screened and copies of appropriate data were made. These data were then keyed into the computer to provide a database of values for compounds for which experimental data are available. The database also served as a basis to check the accuracy of the estimation methods. Upon completion of data collection, estimation of values for the remaining compounds was performed.

The estimates are primarily based on literature methods and empirical procedures developed by the author. If initial analysis using the estimates is favorable, follow-up experimental determination is recommended. This is especially true for highly polar and high molecular weight compounds.

A code of 1 in the tabulation is based on experimental data. A code of 2 indicates an estimate. The estimates with a code of 3 should be considered rough approximations. The estimates with a code of 2 are more accurate. For those substances which undergo decomposition (such as nitro compounds with explosive decomposition and very large compounds with thermal decomposition) prior to attaining the boiling point temperature, the boiling point temperature is a hypothetical value.

A comparison of calculated and data values is shown in Figure 2 for a representative compound. The graph discloses favorable agreement of equation and data.

EXAMPLE

Calculate the vapor pressure of oxygen (O2) at –178.24°C.

Substitution of the Antoine coefficients from the table and temperature into the equation for vapor pressure yields:

$$\log_{10} P = 6.83706 - 339.2095/(-178.24 + 268.70) = 3.08738$$
$$P = 10^{3.08738} = 1222.87 \text{ mmHg}$$

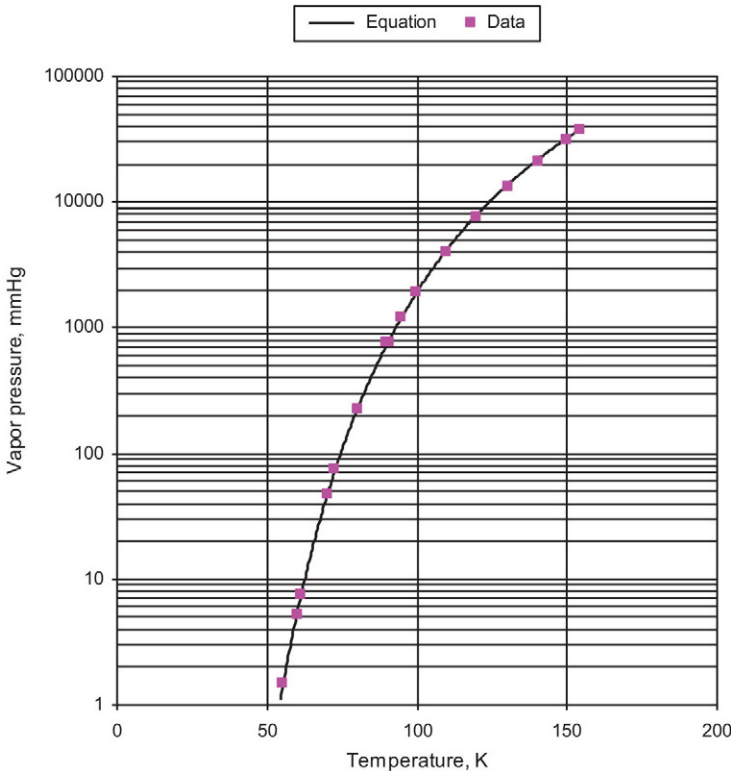


FIGURE 2 Vapor Pressure of Oxygen

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Table 2 Vapor Pressure – Inorganic Compounds

$\log_{10} P = A - B/(T + C)$ (P - mmHg, T - C)									
NO	FORMULA	NAME	CAS No	A	B	C	TMIN	TMAX	code
1	Ac	actinium	7440-34-8	8.85429	20734.8351	273.15	1599.85	3198.00	2
2	Ag	silver	7440-22-4	8.10019	12288.3589	191.93	1009.85	2180.00	1,2
3	AgBr	silver bromide	7785-23-1	6.21353	5179.6010	52.17	569.00	1502.00	1,2
4	AgCl	silver chloride	7783-90-6	8.61748	10917.5071	356.11	912.00	1547.00	1,2
5	AgI	silver iodide	7783-96-2	6.93625	6695.6629	145.03	820.00	1506.00	1,2
6	Al	aluminum	7429-90-5	8.24073	14524.8870	192.25	1209.00	2517.66	1,2
7	AlB3H12	aluminum borohydride	16962-07-5	7.55341	1419.8359	259.36	-52.20	44.50	1,2
8	AlBr3	aluminum bromide	7727-15-3	6.73909	1552.6852	147.43	81.30	255.00	1,2
9	AlCl3	aluminum chloride	7446-70-0	15.75427	5649.0504	258.81	100.00	180.00	1,2
10	AlF3	aluminum fluoride	7784-18-1	12.51245	14983.2501	279.63	744.00	1276.00	1,2
11	AlI3	aluminum iodide	7784-23-8	6.93225	2000.0426	111.66	178.00	382.00	1,2
12	Al2O3	aluminum oxide	1344-28-1	10.96973	25616.3739	186.85	2147.95	2980.00	1,2
13	Al2S3O12	aluminum sulfate	10043-01-3	-0.14253	651.9161	-975.34	572.00	770.05	1,2
14	Am	americium	7440-35-9	9.46480	14359.6923	273.15	965.85	1082.85	1,2
15	Ar	argon	7440-37-1	6.84064	340.2707	271.80	-189.37	-122.29	1,2
16	As	arsenic	7440-38-3	10.60233	6313.8843	216.09	280.00	601.00	1,2
17	AsBr3	arsenic bromide	7784-33-0	7.22542	1953.4063	228.62	41.80	221.00	1,2
18	AsCl3	arsenic trichloride	7784-34-1	7.44841	1670.0897	235.64	-11.40	130.00	1,2
19	AsF3	arsenic trifluoride	7784-35-2	7.30253	1189.5299	211.22	-2.50	57.80	1,2
20	AsF5	arsenic pentafluoride	7784-36-3	5.74019	366.9598	181.14	-117.90	-52.80	1,2
21	AsH3	arsine	7784-42-1	6.88872	790.0472	259.60	-116.92	99.85	1,2
22	AsI3	arsenic triiodide	7784-45-4	7.30188	2551.7378	209.53	164.00	329.50	1,2
23	As2O3	arsenic trioxide	1327-53-3	4.99289	1001.4246	6.99	133.70	467.15	1,2
24	At	astatine	7440-68-8	9.64649	3910.8049	244.23	88.00	259.00	1,2
25	Au	gold	7440-57-5	8.16676	16053.2805	180.98	952.85	2856.00	1,2
26	B	boron	7440-42-8	9.44507	26209.9893	190.34	2075.00	3799.00	1,2
27	BBr3	boron tribromide	10294-33-4	6.58372	1114.5997	210.01	-45.00	91.00	1,2
28	BCl3	boron trichloride	10294-34-5	6.82139	972.2015	234.22	-107.00	178.80	1,2
29	BF3	boron trifluoride	7637-07-2	7.29112	618.3678	240.01	-128.37	-12.25	1,2
30	BH2CO	borine carbonyl	---	7.47035	893.9306	258.95	-139.20	-64.00	2
31	BI3	boron triiodide	13517-10-7	6.93670	1698.2347	210.58	43.00	200.00	1,2
32	B2D6	diborane, deutero	20396-66-1	6.30635	512.0855	241.99	-154.90	-92.50	1,2
33	B2F4	diborane, tetrafluoro	13965-73-6	7.59073	1126.3767	273.15	-40.00	-14.00	1,2
34	B2H5Br	diborane hydrobromide	---	7.79587	1454.1004	279.55	-93.30	16.30	1,2
35	B2H6	diborane	19287-45-7	6.83515	654.3293	257.97	-165.50	16.65	1,2
36	B3H6N3	borazine	6569-51-3	7.19573	1254.5863	237.76	-63.00	53.00	1,2
37	B4H10	tetraborane	18283-93-7	7.23266	1190.0772	255.46	-90.90	18.00	1,2
38	B5H9	pentaborane	19624-22-7	6.46328	948.6930	204.82	-40.40	295.30	1,2
39	B5H11	pentaborane	18433-84-6	9.78620	2656.8429	321.75	-50.20	63.00	1,2
40	B10H14	decaborane	17702-41-9	4.60815	512.9508	48.92	60.00	163.80	1,2
41	Ba	barium	7440-39-3	6.47688	7779.9111	266.45	638.00	1897.00	1,2
42	BaF2	barium fluoride	7787-32-8	8.20759	12116.9768	20.38	1368.00	2250.00	1,2
43	BaO	barium oxide	1304-28-5	12.51502	21900.0000	273.15	1950.00	2000.00	1,2
44	Be	beryllium	7440-41-7	8.08396	13696.6102	148.52	823.85	2483.85	1,2
45	BeB2H8	beryllium borohydride	17440-85-6	15.84584	6333.8132	398.57	1.00	90.00	1,2
46	BeBr2	beryllium bromide	7787-46-4	11.58765	6557.5715	274.67	289.00	474.00	1,2
47	BeCl2	beryllium chloride	7787-47-5	11.21338	6220.1613	264.49	291.00	482.00	1,2
48	BeF2	beryllium fluoride	7787-49-7	9.03258	7534.4285	55.76	686.00	1169.00	1,2
49	BeI2	beryllium iodide	7787-53-3	10.16571	5302.6269	240.89	283.00	487.00	1,2
50	Bi	bismuth	7440-69-9	7.90580	9022.0737	231.44	668.00	1564.00	1,2
51	BiBr3	bismuth tribromide	7787-58-8	8.73564	4109.4999	248.90	261.00	453.00	1,2
52	BiCl3	bismuth trichloride	7787-60-2	7.31869	2847.8317	194.71	230.50	447.00	1,2
53	BrFO3	perbromyl fluoride	25251-03-0	7.43651	1195.8000	260.10	-75.15	28.85	1,2
54	BrF3	bromine trifluoride	7787-71-5	7.72974	1673.9500	219.48	25.85	157.85	1,2
55	BrF5	bromine pentafluoride	7789-30-2	7.85138	1488.6143	258.99	-69.30	40.50	1,2
56	Br2	bromine	7726-95-6	7.27906	1321.8955	241.80	-7.30	311.00	1,2
57	C	graphite	7782-42-5	13.52595	41703.0146	280.60	2565.85	3634.85	1,2
58	CCl2O	carbonyl chloride	75-44-5	7.14063	1063.6278	242.13	-127.78	181.85	1,2
59	CF2O	carbonyl fluoride	353-50-4	7.41143	765.5881	252.08	-111.26	23.85	1,2
60	CH4N2O	urea	57-13-6	10.45495	4591.7946	273.76	67.50	94.90	1,2
61	CH4N2S	thiourea	62-56-6	7.57639	2173.4632	198.99	181.00	580.85	2
62	CNBr	cyanogen bromide	506-68-3	6.36918	722.7494	142.72	-0.14	39.94	1,2
63	CNCl	cyanogen chloride	506-77-4	7.55525	1251.9891	254.99	-6.50	175.85	1,2
64	CNF	cyanogen fluoride	1495-50-7	6.75643	689.9197	224.02	-76.40	-46.00	1,2
65	CO	carbon monoxide	630-08-0	6.72527	295.2279	268.24	-205.00	-140.23	1,2
66	COS	carbonyl sulfide	463-58-1	7.00667	833.0673	252.06	-138.80	105.65	1,2
67	COSe	carbon oxyselenide	1603-84-5	7.17236	1024.3828	260.38	-120.00	-22.00	1,2
68	CO2	carbon dioxide	124-38-9	7.58828	861.8205	271.88	-56.57	31.04	1,2
69	CS2	carbon disulfide	75-15-0	7.21790	1303.7883	254.39	-111.57	278.85	1,2
70	C2N2	cyanogen	460-19-5	7.04339	770.7952	206.27	-95.80	-21.10	1,2
71	C3S2	carbon subsulfide	627-34-9	7.51838	1986.8195	250.50	14.00	130.80	2
72	Ca	calcium	7440-70-2	7.10583	6959.2899	162.92	591.00	1482.00	1,2
73	CaCl2	calcium chloride	10043-52-4	9.22000	13570.0000	273.15	836.85	1007.85	1,2
74	CaF2	calcium fluoride	7789-75-5	7.47685	10678.0076	-210.04	1417.85	2533.35	1,2
75	Cd	cadmium	7440-43-9	7.66023	4757.6158	228.44	119.85	767.00	1,2
76	CdBr2	cadmium bromide	7789-42-6	10.71437	9463.5051	364.07	373.00	844.00	1,2
77	CdCl2	cadmium chloride	10108-64-2	9.79511	9774.8935	453.72	618.00	960.00	1,2
78	CdF2	cadmium fluoride	7790-79-6	8.15105	9504.5987	55.45	1112.00	1748.00	1,2
79	CdI2	cadmium iodide	7790-80-9	7.72916	4256.7258	135.98	296.00	742.00	1,2
80	CdO	cadmium oxide	1306-19-0	9.13973	11107.3677	215.65	1000.00	1559.00	1,2

Table 2 Vapor Pressure – Inorganic Compounds

$\log_{10} P = A - B/(T + C)$ (P - mmHg, T - C)

NO	FORMULA	NAME	CAS No	A	B	C	TMIN	TMAX	code
81	CdS	cadmium sulfide	1306-23-6	10.87780	10021.4278	273.15	930.00	980.00	1,2
82	Ce	cerium	7440-45-1	8.54474	20813.1074	231.68	1719.00	3443.00	1,2
83	CeBr3	cerium bromide	14457-87-5	11.23239	14449.4770	273.15	1407.00	1457.00	1,2
84	CeCl3	cerium chloride	7790-86-5	10.38288	15005.2628	273.15	1677.00	1727.00	1,2
85	ClF	chlorine monofluoride	7790-89-8	6.90617	644.1798	249.77	-158.15	26.85	1,2
86	ClFO2	chloryl fluoride	13637-83-7	6.67696	809.7800	218.96	-82.15	21.85	1,2
87	ClFO3	perchloryl fluoride	7616-94-6	7.08309	854.6155	250.03	-147.74	95.25	1,2
88	ClF3	chlorine trifluoride	7790-91-2	7.34481	1086.4081	231.62	-73.70	11.75	1,2
89	ClF5	chlorine pentafluoride	13637-63-3	6.87794	896.9883	237.51	-79.20	24.80	1,2
90	ClNO2	nitryl chloride	13444-90-1	5.37230	395.4000	174.00	-87.15	-21.15	1,2
91	ClO2	chlorine dioxide	10049-04-4	7.29962	1033.2300	222.93	-59.60	191.85	1,2
92	Cl2	chlorine	7782-50-5	7.06306	906.7031	250.83	-101.03	144.00	1,2
93	Cl2O	chlorine monoxide	7791-21-1	7.70028	1296.5974	266.83	-98.50	2.20	1,2
94	Cl2O4	chlorine perchlorate	27218-16-2	7.53867	1404.1800	257.00	-44.15	74.85	1,2
95	Cl2O6	chlorine hexoxide	12442-63-6	5.83238	1400.9600	273.15	151.50	201.50	1,2
96	Cl2O7	chlorine heptoxide	10294-48-1	7.59797	1583.6401	253.72	-45.30	82.00	1,2
97	Co	cobalt	7440-48-4	8.85240	18477.2528	166.22	1517.00	2925.00	1,2
98	CoCl2	cobalt chloride	7646-79-9	7.04947	4946.1939	137.52	770.00	1049.00	1,2
99	CoNC3O4	cobalt nitrosyl tricarbonyl	14096-82-3	3.05780	9.5159	6.92	-1.30	80.00	1,2
100	Cr	chromium	7440-47-3	8.48706	15307.9218	59.51	1383.00	2671.00	1,2
101	CrC6O6	chromium carbonyl	13007-92-6	10.71935	3410.1719	280.20	36.00	151.00	1,2
102	CrCl2	chromium chloride	10049-05-5	12.16568	12935.2174	273.15	1070.00	1120.00	1,2
103	CrCl3	chromium chloride	10025-73-7	10.96666	12720.2532	273.15	1250.00	1300.00	1,2
104	CrO2Cl2	chromium oxychloride	14977-61-8	6.76400	1231.6008	200.16	-18.40	117.00	1,2
105	CrO3	chromium oxide	1333-82-0	20.14000	10300.0000	273.15	174.85	194.85	1,2
106	Cs	cesium	7440-46-2	6.67520	3515.7582	255.57	21.85	671.00	1,2
107	CsBr	cesium bromide	7787-69-1	7.70150	7132.1044	179.48	748.00	1300.00	1,2
108	CsCl	cesium chloride	7647-17-8	7.74073	7233.7811	191.46	744.00	1297.00	1,2
109	CsF	cesium fluoride	13400-13-0	7.54319	6579.0903	160.10	712.00	1251.00	1,2
110	CsI	cesium iodide	7789-17-5	7.64311	6849.9814	158.38	738.00	1280.00	1,2
111	Cu	copper	7440-50-8	8.30991	15006.7869	202.14	1216.00	2562.00	1,2
112	CuBr	cuprous bromide	7787-70-4	5.01163	2858.6991	-3.41	572.00	1345.00	1,2
113	CuCl	cuprous chloride	7758-89-6	4.45573	2327.4421	-12.18	429.85	1490.00	1,2
114	CuCl2	cupric chloride	7447-39-4	-4.76869	3645.4156	-999.92	309.70	521.00	1,2
115	CuI	copper iodide	7681-65-4	4.81291	2646.3848	33.70	610.00	1336.00	1,2
116	DBr	deuterium bromide	13536-59-9	6.16238	505.6800	220.60	-88.15	-51.15	1,2
117	DCN	deuterium cyanide	3017-23-0	6.25464	689.1316	178.31	-68.90	26.20	2
118	DCl	deuterium chloride	7698-05-7	6.93596	668.2000	249.50	-115.15	-62.15	1,2
119	DF	deuterium fluoride	14333-26-7	7.21704	1268.3700	273.87	-84.15	49.85	1,2
120	DI	deuterium iodide	14104-45-1	5.60474	414.6800	187.87	-55.15	-7.15	1,2
121	D2	deuterium	7782-39-0	6.31539	92.2133	276.35	-254.42	-234.80	1,2
122	D2O	deuterium oxide	7789-20-0	8.06347	1708.9717	228.35	3.82	370.74	1,2
123	Dy	dysprosium	7429-91-6	6.67690	9758.5876	3.70	1105.00	2567.00	1,2
124	Er	erbium	7440-52-0	7.07492	12618.2924	140.57	1231.00	2868.00	1,2
125	Eu	europium	7440-53-1	6.80515	6548.9117	143.35	590.00	1523.00	1,2
126	FHO3S	fluorosulfonic acid	7789-21-1	7.39950	1521.0000	174.00	64.85	192.85	1,2
127	FNO	nitrosyl fluoride	7789-25-5	7.88277	986.5042	257.12	-132.00	-59.90	1,2
128	FNO2	nitryl fluoride	10022-50-1	6.94723	685.0522	240.87	-156.00	-72.40	1,2
129	FNO3	fluorine nitrate	7789-26-6	6.53787	721.4837	243.29	-160.00	-46.00	1,2
130	F2	fluorine	7782-41-4	6.89461	320.1001	267.95	-219.67	-128.84	1,2
131	F2O	fluorine oxide	7783-41-7	7.29316	580.8480	276.45	-198.15	-63.15	1,2
132	F2O2	difluorine dioxide	7783-44-0	6.77902	756.3900	250.16	-125.15	-80.15	1,2
133	Fe	iron	7439-89-6	8.37930	16213.5121	88.52	1435.00	2860.20	1,2
134	FeC5O5	iron pentacarbonyl	13463-40-6	9.24491	2718.2717	324.13	-6.50	103.00	1,2
135	FeCl2	ferrous chloride	7758-94-3	7.23943	4667.3502	46.98	700.00	1023.85	1,2
136	FeCl3	ferric chloride	7705-08-0	28.93093	32484.7863	928.01	108.00	319.00	1,2
137	FeI2	iron iodide	7783-86-0	7.09897	4398.3915	107.58	587.00	926.85	1,2
138	Fr	francium	7440-73-5	6.28464	3099.3568	237.55	121.00	673.00	2
139	Ga	gallium	7440-55-3	8.24091	13438.1916	259.39	1037.00	2245.00	1,2
140	GaBr3	gallium bromide	13450-88-9	7.64043	2477.2199	205.34	121.50	226.85	1,2
141	GaCl3	gallium trichloride	13450-90-3	7.54055	2005.3675	229.36	77.75	420.85	1,2
142	Gd	gadolinium	7440-54-2	8.09461	18125.4084	210.60	1563.00	3262.00	1,2
143	Ge	germanium	7440-56-4	8.39437	16953.3407	240.64	1371.00	2833.00	1,2
144	GeBr4	germanium bromide	13450-92-5	6.92749	1650.1032	221.42	43.30	186.35	1,2
145	GeCl4	germanium chloride	10038-98-9	6.27456	971.2957	199.65	-45.00	86.55	1,2
146	GeHCl3	germane trichloro	1184-65-2	7.29569	1304.6359	220.21	-41.30	75.30	1,2
147	GeH4	germane	7782-65-2	7.08333	732.4097	262.43	-165.89	34.85	1,2
148	GeS	germanium sulfide	12025-32-0	6.81801	6765.5528	214.38	615.00	726.85	1,2
149	Ge2H6	digermane	13818-89-8	6.98641	1189.4193	258.91	-88.70	30.80	1,2
150	Ge3H8	trigermane	14691-44-2	7.19898	1588.2499	257.31	-36.90	110.50	1,2
151	HBr	hydrogen bromide	10035-10-6	6.93857	704.4329	240.30	-88.00	90.00	1,2
152	HCN	hydrogen cyanide	74-90-8	8.08422	1655.8693	292.23	-13.32	183.50	1,2
153	HCl	hydrogen chloride	7647-01-0	7.25303	764.8366	259.93	-114.18	51.50	1,2
154	HClO3S	chlorosulfonic acid	7790-94-5	7.47025	1749.2078	227.29	-80.00	426.85	1,2
155	HClO4	perchloric acid	7601-90-3	6.71471	1348.2463	239.81	-101.20	357.85	1,2
156	HF	hydrogen fluoride	7664-39-3	8.12392	1680.6466	301.02	-83.36	188.00	1,2
157	HI	hydrogen iodide	10034-85-2	6.99067	881.4677	250.08	-50.77	150.70	1,2
158	HNO3	nitric acid	7697-37-2	7.56453	1431.7460	222.69	-41.60	102.95	1,2
159	HN3	hydrazoic acid	7782-79-8	6.65340	979.1881	223.46	-79.00	35.70	1,2
160	HPO2F2	difluorophosphoric acid	13779-41-4	6.73530	1342.9000	232.00	4.85	151.85	1,2

Table 2 Vapor Pressure – Inorganic Compounds

$\log_{10} P = A - B/(T + C)$ (P - mmHg, T - C)									
NO	FORMULA	NAME	CAS No	A	B	C	TMIN	TMAX	code
161	H2	hydrogen	1333-74-0	6.14858	80.9480	277.53	-259.20	-239.97	1,2
162	H2O	water	7732-18-5	8.05573	1723.6425	233.08	0.01	373.98	1,2
163	H2O2	hydrogen peroxide	7722-84-1	8.07489	1958.1824	226.80	-0.41	457.00	1,2
164	H2S	hydrogen sulfide	7783-06-4	7.11958	802.2266	249.61	-85.47	100.38	1,2
165	H2SO4	sulfuric acid	7664-93-9	8.28538	2976.8370	213.95	25.00	336.85	1,2
166	H2S2	hydrogen disulfide	13465-07-1	7.02654	1242.2189	228.94	-40.00	70.70	1,2
167	H2S3	dihydrogen trisulfide	13845-23-3	6.80710	1488.0000	209.00	54.85	200.85	1,2
168	H2S4	dihydrogen tetrasulfide	13845-25-5	6.94510	1772.0000	196.00	110.85	273.85	1,2
169	H2S5	dihydrogen pentasulfide	13845-24-4	7.32010	2104.0000	189.00	152.85	318.85	1,2
170	H2Se	hydrogen selenide	7783-07-5	6.46297	587.4538	205.99	-115.30	-42.00	1,2
171	H2Te	hydrogen telluride	7783-09-7	5.90921	586.0450	195.52	-96.40	-2.00	1,2
172	H3NO	hydroxylamine	7803-49-8	11.71366	3155.6880	247.42	33.10	109.85	1,2
173	H3SNO3	sulfamic acid	5329-14-6	0.11312	684.4925	273.15	20.00	100.00	2
174	He	helium-3	14762-55-1	4.82950	8.5576	274.36	-272.15	-269.83	1,2
175	He	helium-4	7440-59-7	5.27120	13.5171	274.58	-271.39	-267.95	1,2
176	Hf	hafnium	7440-58-6	9.06491	30232.9115	285.82	2416.00	4603.00	1,2
177	HfCl4	hafnium chloride	13499-05-3	11.71000	5197.0000	273.15	202.85	310.00	1,2
178	HfI3	hafnium iodide	13779-73-2	19.56000	10700.0000	273.15	301.85	323.85	1,2
179	Hg	mercury	7439-97-6	7.89458	3147.6142	271.10	-38.84	1461.85	1,2
180	HgBr2	mercuric bromide	7789-47-1	5.56804	974.6103	40.68	136.50	322.00	1,2
181	HgCl2	mercuric chloride	7487-94-7	12.19380	6369.7914	379.97	136.20	304.00	1,2
182	HgI2	mercuric iodide	7774-29-0	7.11921	2015.6871	121.58	157.50	354.00	1,2
183	Ho	holmium	7440-60-0	6.66154	10226.4211	4.88	1159.00	2700.00	1,2
184	ICl	iodine chloride	7790-99-0	7.69746	1515.4000	216.80	13.85	123.85	1,2
185	IF5	iodine pentafluoride	7783-66-6	7.24100	1387.0000	213.62	9.85	134.85	1,2
186	IF7	iodine heptafluoride	16921-96-3	8.58999	1562.5974	268.90	-87.00	4.80	1,2
187	I2	iodine	7553-56-2	7.10736	1556.9694	183.97	-31.15	546.00	1,2
188	In	indium	7440-74-6	7.98843	11856.7391	249.38	900.00	2072.00	1,2
189	InBr	indium bromide	14280-53-6	19.02460	15000.0000	273.15	626.00	656.00	1,2
190	InBr3	indium bromide	13465-09-3	5.89173	727.4250	-183.70	304.60	364.80	1,2
191	InCl	indium chloride	13465-10-6	8.03000	4640.0000	273.15	211.00	576.85	1,2
192	InCl3	indium chloride	10025-82-8	13.62000	8270.0000	273.15	226.85	426.85	1,2
193	Ir	iridium	7439-88-5	9.63425	30934.6262	190.68	2440.00	4386.00	1,2
194	IrF6	iridium fluoride	7783-75-7	8.56221	1717.5557	248.71	-88.00	53.60	1,2
195	K	potassium	7440-09-7	6.97431	4143.9271	254.47	63.20	1949.85	1,2
196	KBr	potassium bromide	7758-02-3	7.60549	7291.2713	163.23	795.00	1380.00	1,2
197	KCl	potassium chloride	7447-40-7	6.93314	5810.3508	17.98	770.85	3196.85	1,2
198	KF	potassium fluoride	7789-23-3	7.61993	7726.2515	128.32	885.00	1502.00	1,2
199	KI	potassium iodide	7681-11-0	6.57543	5053.2934	22.90	745.00	1344.85	1,2
200	KOH	potassium hydroxide	1310-58-3	7.08223	6300.4849	172.76	405.85	1326.85	1,2
201	Kr	krypton	7439-90-9	7.01408	511.6161	277.13	-157.37	-63.80	1,2
202	La	lanthanum	7439-91-0	8.57782	21096.9943	239.17	1650.00	3464.00	1,2
203	LaBr3	lanthanum bromide	13536-79-3	10.98826	15000.0000	273.15	1547.00	1577.00	1,2
204	LaCl3	lanthanum chloride	10099-58-8	14.66261	15000.0000	273.15	970.00	1000.00	1,2
205	Li	lithium	7439-93-2	7.80332	7953.8982	276.84	524.30	1337.10	1,2
206	LiBr	lithium bromide	7550-35-8	7.57994	6947.3452	168.43	748.00	1310.00	1,2
207	LiCl	lithium chloride	7447-41-8	7.53439	7299.6503	185.61	783.00	1383.00	1,2
208	LiF	lithium fluoride	7789-24-4	8.71722	11054.9564	221.14	1047.00	1673.00	1,2
209	LiI	lithium iodide	10377-51-2	8.49354	7459.8616	155.25	723.00	1173.85	1,2
210	Lu	lutetium	7439-94-3	7.83211	17422.3777	116.75	1600.00	3402.00	1,2
211	Mg	magnesium	7439-95-4	7.16078	5253.0612	137.71	428.00	1088.00	1,2
212	MgBr2	magnesium bromide	7789-48-2	7.38742	6095.1378	104.02	711.00	1226.85	1,2
213	MgCl2	magnesium chloride	7786-30-3	7.13801	6696.7751	161.05	778.00	1412.00	1,2
214	MgF2	magnesium fluoride	7783-40-6	7.87219	12036.3932	82.67	1262.85	2126.85	1,2
215	MgI2	magnesium iodide	10377-58-9	6.92855	4121.1468	109.39	634.00	876.85	1,2
216	MgO	magnesium oxide	1309-48-4	6.92276	13862.2156	-170.46	2831.85	5676.85	1,2
217	Mn	manganese	7439-96-5	7.50506	9853.7049	68.23	955.00	2060.00	1,2
218	MnCl2	manganese chloride	7773-01-5	7.03063	5467.6217	127.56	736.00	1190.00	1,2
219	MnF2	manganese fluoride	7782-64-1	16.16514	15000.0000	273.15	826.00	856.00	1,2
220	Mo	molybdenum	7439-98-7	8.64938	26943.1312	31.68	2400.00	4950.00	1,2
221	MoCl5	molybdenum chloride	10241-05-1	6.29946	1850.0000	273.15	218.00	268.00	1,2
222	MoF5	molybdenum fluoride	13819-84-6	8.49628	2709.7945	268.96	86.60	213.60	1,2
223	MoF6	molybdenum fluoride	7783-77-9	7.06989	1018.5440	209.14	-65.50	34.00	1,2
224	MoOF4	molybdenum oxytetrafluoride	14459-59-7	8.53528	2492.6355	254.83	-21.00	186.00	1,2
225	MoO3	molybdenum oxide	1313-27-5	6.05246	2898.1602	-233.00	801.00	1140.00	1,2
226	NClO	nitrosyl chloride	2696-92-6	7.83560	1314.8605	270.75	-59.60	167.50	1,2
227	NCl3	nitrogen trichloride	10025-85-1	7.00316	1212.7111	223.18	-27.00	94.00	1,2
228	ND3	heavy ammonia	13550-49-7	6.83776	746.0100	221.53	-74.00	-33.00	1,2
229	NF3	nitrogen trifluoride	7783-54-2	7.01648	559.5184	264.35	-206.79	-39.30	1,2
230	NF3S	nitrogen fluoride sulfide	15930-75-3	6.88510	888.5000	245.00	-94.15	1.85	1,2
231	NH3	ammonia	7664-41-7	7.58743	1013.7815	248.83	-77.74	132.50	1,2
232	NH4Br	ammonium bromide	12124-97-9	9.74168	4588.8568	272.85	198.30	396.00	1,2
233	NH4Cl	ammonium chloride	12125-02-9	8.93371	3215.2979	192.35	280.00	520.00	1,2
234	NH4I	ammonium iodide	12027-06-4	10.40721	5286.5838	297.41	210.90	405.00	1,2
235	NH5O	ammonium hydroxide	1336-21-6	7.47687	1118.7172	245.15	-70.00	80.00	2
236	NH5S	ammonium hydrogensulfide	12124-99-1	8.96559	1598.3103	229.34	-51.10	33.30	2
237	NO	nitric oxide	10102-43-9	7.70299	473.0296	249.87	-163.65	-93.00	1,2
238	NO2	nitrogen dioxide	10102-44-0	9.74325	2289.6342	312.65	-11.20	158.20	1,2
239	N2	nitrogen	7727-37-9	6.72531	285.5727	270.09	-210.00	-147.05	1,2
240	N2F4	tetrafluorohydrazine	10036-47-2	7.41740	952.4783	284.15	-161.50	36.20	1,2

Table 2 Vapor Pressure – Inorganic Compounds

$\log_{10} P = A - B/(T + C)$ (P - mmHg, T - C)

NO	FORMULA	NAME	CAS No	A	B	C	TMIN	TMAX	code
241	N2H4	hydrazine	302-01-2	7.82877	1698.5808	229.79	1.53	380.00	1,2
242	N2H4C	ammonium cyanide	12211-52-8	9.16524	1644.6341	230.00	-50.60	31.70	1,2
243	N2H6CO2	ammonium carbamate	1111-78-0	10.32341	2267.7036	245.93	-26.10	58.30	2
244	N2O	nitrous oxide	10024-97-2	6.81488	550.6045	228.44	-90.85	36.42	1,2
245	N2O3	nitrogen trioxide	10544-73-7	7.60164	1109.3200	232.98	-103.15	151.85	1,2
246	N2O4	nitrogen tetraoxide	10544-72-6	4.09407	97.8287	59.47	-11.25	47.50	1,2
247	N2O5	nitrogen pentoxide	10102-03-1	11.93250	2617.6543	256.19	-36.80	33.00	1,2
248	Na	sodium	7440-23-5	7.33576	5060.9675	254.35	280.60	926.85	1,2
249	NaBr	sodium bromide	7647-15-6	7.54103	7142.9394	142.08	746.85	1446.85	1,2
250	NaCN	sodium cyanide	143-33-9	7.38628	7836.8557	243.41	563.70	1496.00	1,2
251	NaCl	sodium chloride	7647-14-5	7.96318	8432.2350	194.07	800.75	1465.05	1,2
252	NaF	sodium fluoride	7681-49-4	8.61974	11408.3735	278.32	995.85	1786.85	1,2
253	NaI	sodium iodide	7681-82-5	7.89004	7361.9912	165.84	767.00	1303.85	1,2
254	NaOH	sodium hydroxide	1310-73-2	7.46511	7606.4353	280.16	500.00	1600.00	1,2
255	Na2SO4	sodium sulfate	7757-82-6	11.40104	15657.1880	273.15	1514.50	1564.50	2
256	Nb	niobium	7440-03-1	9.92532	35189.3422	251.22	2669.00	4740.00	1,2
257	NbBr5	niobium bromide	13478-45-0	9.33000	4085.0000	273.15	254.00	360.00	1,2
258	NbCl4	niobium chloride	13569-70-5	12.30000	6870.0000	273.15	303.85	377.85	1,2
259	NbCl5	niobium chloride	10026-12-7	8.37000	2870.0000	273.15	204.70	246.85	1,2
260	NbF5	niobium fluoride	7783-68-8	6.88947	1791.5982	217.93	86.30	229.00	1,2
261	Nd	neodymium	7440-00-8	6.95222	12884.3059	97.12	1322.30	3063.00	1,2
262	NdBr3	neodymium tribromide	13536-80-6	11.15371	15000.0000	273.15	1510.00	1540.00	1,2
263	NdCl3	neodymium trichloride	10024-93-8	10.88871	15000.0000	273.15	1570.00	1600.00	1,2
264	Ne	neon	7440-01-9	6.74431	115.3285	275.91	-248.59	-228.75	1,2
265	Ni	nickel	7440-02-0	8.74585	17882.3827	134.99	1510.00	2911.00	1,2
266	NiC4O4	nickel carbonyl	13463-39-3	7.64987	1479.9228	267.82	-23.00	42.50	1,2
267	NiCl2	nickel chloride	7718-54-9	10.25227	8220.2197	130.14	534.00	985.00	1,2
268	NiF2	nickel fluoride	10028-18-9	8.31881	7276.8348	-160.62	1077.00	1283.00	2
269	Np	neptunium	7439-99-8	7.90387	22003.2016	273.15	1726.85	2226.85	2
270	NpF6	neptunium hexafluoride	14521-05-2	7.19634	1271.8225	239.53	54.85	76.85	1,2
271	O2	oxygen	7782-44-7	6.83706	339.2095	268.70	-218.80	-118.57	1,2
272	O3	ozone	10028-15-6	7.15980	618.6072	255.87	-193.00	-12.15	1,2
273	Os	osmium	7440-04-2	9.76546	34751.0553	35.61	2600.00	5400.00	1,2
274	OsF5	osmium fluoride	31576-40-6	9.75828	3432.7793	273.23	74.10	225.90	1,2
275	OsF6	osmium fluoride	13768-38-2	8.26547	1525.6447	235.83	-89.00	47.50	1,2
276	OsOF5	osmium oxide pentafluoride	---	3.06277	47.1535	1.04	31.65	57.75	2
277	OsO4	osmium tetroxide	20816-12-0	4.72185	404.5748	84.75	3.20	135.00	1,2
278	P	phosphorus (white)	7723-14-0	7.00000	2016.5981	215.00	69.00	317.00	1,2
279	P	phosphorus (red)	7723-14-0	11.39619	5737.7187	242.50	181.85	431.00	1,2
280	PBrF2	phosphorus bromide difluoride	15597-40-7	6.90420	885.1200	236.00	-91.15	10.85	1,2
281	PBr2F	phosphorus dibromide fluoride	15597-39-4	6.85800	1210.3000	226.00	-18.15	109.85	1,2
282	PBr3	phosphorus tribromide	7789-60-8	7.35097	1883.6717	248.44	7.80	172.95	1,2
283	PBr5	phosphorus bromide	7789-69-7	10.31441	2746.6473	259.11	-19.00	110.10	1,2
284	PClF2	phosphorus chloride difluoride	14335-40-1	7.34733	1041.3236	280.39	-119.50	89.25	1,2
285	PCl2F	phosphorus dichloride fluoride	15597-63-4	7.11413	1120.6064	250.71	-71.10	189.85	1,2
286	PCl2F3	phosphorus dichloride trifluoride	13454-99-4	7.38018	1175.5119	257.82	-79.80	-22.80	1,2
287	PCl3	phosphorus chloride	7719-12-2	6.87188	1218.9462	229.32	-92.00	101.00	1,2
288	PCl5	phosphorus pentachloride	10026-13-8	10.28827	2973.0094	241.50	-12.00	159.85	1,2
289	PF3	phosphorus fluoride	7783-55-3	7.06415	651.8743	257.33	-152.00	-1.95	1,2
290	PF3O	phosphorus oxyfluoride	13478-20-1	10.28247	1549.5473	248.89	-124.00	-39.70	1,2
291	PF5	phosphorus fluoride	7647-19-0	9.61509	1143.4832	254.40	-157.00	-84.60	1,2
292	PH3	phosphine	7803-51-2	6.98375	724.4708	264.31	-133.78	51.60	1,2
293	PH4Br	phosphonium bromide	---	10.09866	2085.9953	250.37	-43.70	38.30	2
294	PH4Cl	phosphonium chloride	24567-53-1	9.08492	1255.1903	229.31	-91.00	-27.00	2
295	PH4I	phosphonium iodide	12125-09-6	10.23174	2291.9882	249.30	-25.20	62.50	1,2
296	POBrClF	phosphoryl bromide chloride fluoride	14518-81-1	6.91410	1214.0000	222.00	-10.15	102.85	1,2
297	POBrCl2	phosphoryl bromide dichloride	13455-03-3	6.92410	1411.0000	213.00	31.85	162.85	1,2
298	POBrF2	phosphoryl bromide difluoride	14014-18-7	7.10190	1118.9000	233.00	-44.15	51.85	1,2
299	POBr3	phosphorus oxybromide	7789-59-5	7.13796	1697.4936	207.04	64.00	191.70	1,2
300	POClF2	phosphoryl chloride difluoride	13769-75-0	6.92660	946.9600	231.00	-76.15	30.85	1,2
301	POCl2F	phosphoryl dichloride fluoride	13769-76-1	7.08465	1201.8600	233.00	-40.15	84.85	1,2
302	POCl3	phosphorus oxychloride	10025-87-3	7.05109	1390.0324	227.82	1.18	105.50	1,2
303	PSBr3	phosphorus thiobromide	3931-89-3	9.40218	2659.4144	232.93	50.00	175.00	2
304	PSCl3	phosphorus thiochloride	3982-91-0	7.11462	1501.0578	229.54	-36.20	125.00	1,2
305	P2O3	phosphorus trioxide	1314-24-5	8.39335	2597.3392	298.17	47.30	173.00	1,2
306	P2O5	phosphorus pentoxide	1314-56-3	10.74529	6470.4817	217.75	230.00	700.00	1,2
307	P2S5	phosphorus sulfide	1314-80-3	9.02309	4959.9962	293.52	288.00	1017.85	1,2
308	P4O6	phosphorous oxide	12440-00-5	8.35860	2568.0727	295.82	39.70	173.00	1,2
309	Pb	lead	7439-92-1	7.46345	9131.8353	243.70	434.85	1749.00	1,2
310	PbBr2	lead bromide	10031-22-8	8.87105	6987.2446	274.44	513.00	892.00	1,2
311	PbCl2	lead chloride	7758-95-4	7.64617	5101.5890	119.56	547.00	951.00	1,2
312	PbF2	lead fluoride	7783-46-2	7.62479	6525.7272	82.58	861.00	1293.00	1,2
313	PbI2	lead iodide	10101-63-0	6.76027	3580.4550	50.32	479.00	872.00	1,2
314	PbO	lead oxide	1317-36-8	9.24183	10820.2993	227.03	943.00	1474.00	1,2
315	PbO	lead oxide (massicot)	1317-36-8	9.33076	11048.8015	240.48	724.00	1471.00	1,2
316	PbS	lead sulfide	1314-87-0	8.84745	7859.9264	36.31	852.00	1281.00	1,2
317	Pd	palladium	7440-05-3	8.56766	18403.4082	273.13	1300.00	3400.00	1,2
318	Po	polonium	7440-08-6	7.18388	5265.0104	261.55	500.00	1050.00	1,2
319	PoCl4	polonium tetrachloride	10026-02-5	9.33621	4280.8340	273.15	300.60	410.00	1,2
320	Pr	praseodymium	7440-10-0	6.78062	14030.7383	77.80	1497.70	3520.00	1,2

Table 2 Vapor Pressure – Inorganic Compounds

$\log_{10} P = A - B/(T + C)$ (P - mmHg, T - C)									
NO	FORMULA	NAME	CAS No	A	B	C	TMIN	TMAX	code
321	PrCl3	praseodymium chloride	10361-79-2	10.44454	15000.0000	273.15	1680.00	1710.00	1,2
322	Pt	platinum	7440-06-4	8.58584	22055.7478	41.02	2057.00	3825.00	1,2
323	Pu	plutonium	7440-07-5	8.08674	18531.9774	331.78	1483.00	3228.00	1,2
324	PuBr3	plutonium bromide	15752-46-2	8.15259	8352.8990	93.61	681.00	1476.85	1,2
325	PuCl3	plutonium chloride	13569-62-5	7.22981	8007.1303	48.46	760.00	1776.85	1,2
326	PuF3	plutonium fluoride	13842-83-6	9.39122	14145.3938	49.57	1396.00	1726.85	1,2
327	Ra	radium	7440-14-4	6.71596	6622.4566	200.78	546.00	1526.00	1,2
328	Rb	rubidium	7440-17-7	6.99992	3926.1751	265.16	160.40	688.00	1,2
329	RbBr	rubidium bromide	7789-39-1	8.13532	8288.1158	237.34	781.00	1340.00	1,2
330	RbCl	rubidium chloride	7791-11-9	7.05736	6116.0391	74.38	792.00	1390.00	1,2
331	RbF	rubidium fluoride	13446-74-7	6.21681	3433.9886	-380.62	921.00	1410.00	1,2
332	RbI	rubidium iodide	7790-29-6	7.66815	7039.8460	170.51	748.00	1300.00	1,2
333	Re	rhenium	7440-15-5	8.97642	34838.4703	119.34	3030.00	5681.00	1,2
334	ReF5	rhenium fluoride	30937-52-1	8.98978	3010.3083	271.47	58.80	221.30	1,2
335	ReF6	rhenium fluoride	10049-17-9	8.47754	1550.5300	243.24	-97.00	33.80	1,2
336	ReF7	rhenium fluoride	17029-21-9	11.47517	5114.2585	521.55	48.30	71.85	1,2
337	ReOF4	rhenium oxytetrafluoride	17026-29-8	9.45214	2533.4840	213.84	5.00	171.70	1,2
338	ReOF5	rhenium oxypentafluoride	23377-53-9	7.90580	1772.1930	279.68	-103.00	73.00	1,2
339	ReO2F3	rhenium dioxytrifluoride	57246-89-6	10.36000	3437.0000	273.15	90.00	181.85	1,2
340	Re2O7	rhenium heptoxide	1314-68-7	10.48807	4082.7928	176.70	120.00	500.00	1,2
341	Rh	rhodium	7440-16-6	9.68520	26833.7062	248.59	2015.00	3695.00	1,2
342	Rn	radon	10043-92-2	6.58011	651.6732	237.86	-163.00	103.85	1,2
343	Ru	ruthenium	7440-18-8	9.33600	26794.4200	33.09	2315.00	4115.00	1,2
344	RuF5	ruthenium fluoride	14521-18-7	7.56315	2037.5032	196.96	49.60	156.80	1,2
345	S	sulfur	7704-34-9	7.86630	3464.9835	250.34	102.00	444.00	1,2
346	SCl2	sulfur dichloride	10545-99-0	8.36643	1559.0062	224.60	-76.00	59.60	1,2
347	SFN	thiazyl fluoride	18820-63-8	6.48180	877.1000	239.00	-8.15	29.85	1,2
348	SF2N2	dinitrogen sulfur difluoride	500010-01-5	6.78280	901.0000	242.00	-83.15	8.85	1,2
349	SF4	sulfur tetrafluoride	7783-60-0	7.02242	882.8300	253.61	-100.00	-40.45	1,2
350	SF6	sulfur hexafluoride	2551-62-4	6.87699	676.2391	233.12	-168.00	45.54	1,2
351	SOBr2	thionyl bromide	507-16-4	8.17193	2202.4959	276.26	-6.70	140.00	1,2
352	SOClF	thionyl chloride fluoride	14177-25-4	7.17310	1100.1000	244.00	-65.15	36.85	1,2
353	SOCl2	thionyl chloride	7719-09-7	7.27960	1415.8028	246.26	-101.15	99.00	1,2
354	SOF2	thionyl fluoride	7783-42-8	6.28388	569.2883	211.09	-98.73	-43.80	1,2
355	SOF4	sulfur oxide tetrafluoride	13709-54-1	7.07180	840.3000	249.00	-113.15	-25.15	1,2
356	SO2	sulfur dioxide	7446-09-5	7.33311	1013.4609	237.65	-75.48	157.60	1,2
357	SO2BrF	sulfuryl bromide fluoride	13536-61-3	7.14280	1155.0000	231.00	-45.15	67.85	1,2
358	SO2ClF	sulfuryl chloride fluoride	13637-84-8	6.52150	793.7300	210.70	-72.15	36.85	1,2
359	SO2Cl2	sulfuryl chloride	7791-25-5	6.94482	1194.8068	224.60	-51.15	271.85	1,2
360	SO2F2	sulfuryl fluoride	2699-79-8	6.90700	784.3000	250.00	-121.15	-32.15	1,2
361	SO3	sulfur trioxide	7446-11-9	6.85638	754.8178	145.11	16.80	217.70	1,2
362	S2Br2	sulfur bromide	13172-31-1	7.17834	1664.6387	185.93	-7.00	200.90	1,2
363	S2Cl2	sulfur monochloride	10025-67-9	7.04178	1466.8804	215.53	-7.40	137.00	1,2
364	S2F2	disulfur difluoride	16860-99-4	6.68410	628.0000	256.00	-120.15	-77.15	1,2
365	S2F10	sulfur decafluoride	5714-22-7	7.06760	1100.6000	234.00	-47.15	48.85	1,2
366	S2O2F10	thiosulfuryl decafluoride	12395-41-4	6.87410	1110.0000	229.00	-34.15	70.85	1,2
367	S2O5ClF	pyrosulfuryl chloride fluoride	13637-85-9	7.01560	1257.4000	204.00	4.85	130.85	1,2
368	S2O5Cl2	pyrosulfuryl chloride	7791-27-7	7.01910	1460.0000	202.00	47.85	177.85	1,2
369	S2O5F2	pyrosulfuryl difluoride	13036-75-4	6.88110	1120.0000	229.00	-33.15	72.85	1,2
370	S2O5F4	disulfur pentoxide tetrafluoride	44982-62-9	6.88510	1140.0000	227.00	-27.15	79.85	1,2
371	S3O8F2	trisulfur octoxide difluoride	13709-33-6	6.99010	1360.0000	211.00	22.85	145.85	1,2
372	Sb	antimony	7440-36-0	5.88125	4984.6752	74.32	534.00	1587.00	1,2
373	SbBr3	antimony tribromide	7789-61-9	8.00960	2654.1840	237.51	93.90	280.00	1,2
374	SbCl3	antimony trichloride	10025-91-9	6.91949	1709.1498	202.95	73.40	520.85	1,2
375	SbCl5	antimony pentachloride	7647-18-9	7.76515	2046.6707	240.86	22.70	114.10	1,2
376	SbH3	stibine	7803-52-3	7.20112	1062.6666	263.97	-95.28	167.20	1,2
377	SbI3	antimony triiodide	7790-44-5	7.55355	2915.0705	222.85	163.60	401.00	1,2
378	Sb2O3	antimony oxide	1309-64-4	4.06652	1464.9462	-189.49	400.00	1550.00	1,2
379	Sc	scandium	7440-20-2	7.72648	13961.2550	45.18	1300.00	3200.00	1,2
380	ScBr3	scandium bromide	13465-59-3	14.35000	13780.0000	273.15	768.85	926.85	1,2
381	ScCl3	scandium chloride	10361-84-9	14.37000	14200.0000	273.15	791.85	959.85	1,2
382	ScI3	scandium iodide	14474-33-0	14.17000	13340.0000	273.15	736.85	906.85	1,2
383	Se	selenium (gray)	7782-49-2	7.05403	3441.9628	139.77	227.00	685.00	1,2
384	SeCS	carbonyl selenide sulfide	5951-19-9	6.89093	1274.4371	232.21	-47.30	85.60	1,2
385	SeCl4	selenium tetrachloride	10026-03-6	10.79338	3477.8728	248.14	74.00	191.40	1,2
386	SeF4	selenium tetrafluoride	13465-66-2	7.61816	1471.6008	204.64	13.60	106.00	1,2
387	SeF6	selenium hexafluoride	7783-79-1	6.99021	727.3425	222.79	-153.00	72.35	1,2
388	SeOCl2	selenium oxychloride	7791-23-3	7.90109	1958.4938	213.12	34.80	177.00	1,2
389	SeOF2	selenium oxyfluoride	7783-43-9	7.42010	1380.0000	178.00	36.85	152.85	1,2
390	SeO2	selenium dioxide	7446-08-4	18.01392	11598.1403	451.41	115.00	335.00	1,2
391	Si	silicon	7440-21-3	8.71307	20614.1581	267.17	1634.85	3264.85	1,2
392	SiBrCl2F	silane bromodichlorofluoro	---	7.51839	1472.7873	282.18	-86.50	201.15	1,2
393	SiBr2ClF	silane dibromochlorofluoro	---	7.26677	1379.5487	255.04	-65.20	242.77	1,2
394	SiCCl3N	silane trichloro-carbonitrile	18157-01-2	7.75100	1687.0000	273.15	-46.15	73.24	1,2
395	SiCCl3NS	silane trichloroisothio-cyanato	18157-00-1	7.99850	2060.0000	273.15	66.85	129.38	1,2
396	SiCH2Cl4O	silane trichloro(chloro-methoxy)	18157-08-9	8.39500	2159.0000	273.15	-0.15	118.39	1,2
397	SiCH3Cl2F	silane dichlorofluoro-methyl	420-58-6	7.74530	1472.3000	273.15	-30.15	-27.15	1,2
398	SiCH3Cl3	silane methyltrichloro	75-79-6	7.18818	1370.6632	251.81	-77.80	243.85	1,2
399	SiCH3F3	silane trifluoromethyl	373-74-0	7.97520	1238.1000	273.15	-60.15	-57.15	1,2
400	SiCH4ClF	silane chlorofluoro-methyl	373-67-1	7.60660	1296.6000	273.15	-75.15	1.22	1,2

Table 2 Vapor Pressure – Inorganic Compounds

$\log_{10} P = A - B/(T + C)$ (P - mmHg, T - C)

NO	FORMULA	NAME	CAS No	A	B	C	TMIN	TMAX	code
401	SiCH4Cl2	silane methyl dichloro	75-54-7	7.23413	1272.2113	250.69	-90.60	209.85	1,2
402	SiCH4F2	silane difluoromethyl	420-34-8	7.76550	1160.4000	273.15	-75.15	-35.59	1,2
403	SiCH5Cl	silane methyl chloro	993-00-0	7.04725	1054.7751	244.46	-134.10	168.85	1,2
404	SiCH5Cl	silane (chloromethyl)	10112-09-1	8.09302	1719.2800	298.60	-27.15	31.26	1,2
405	SiCH6	silane methyl	992-94-9	7.10688	873.3901	263.57	-156.81	79.35	1,2
406	SiCH6O	methyl silyl ether	2171-96-2	3.98012	185.6317	149.98	-90.20	18.88	1,2
407	SiC2H8	silane dimethyl	1111-74-6	6.73312	853.3608	241.12	-150.22	128.85	1,2
408	SiC3H10	silane trimethyl	993-07-7	6.90193	1030.4100	249.55	-135.89	158.85	1,2
409	SiC4H12	silane tetramethyl	75-76-3	6.87190	1060.6645	239.11	-99.08	177.25	1,2
410	SiClF3	silane chlorotrifluoro	14049-36-6	7.06315	761.3693	252.04	-144.00	35.68	1,2
411	SiCl2F2	silane dichlorodifluoro	18356-71-3	7.33305	1054.7676	268.91	-124.70	94.20	1,2
412	SiCl3F	silane trichlorofluoro	14965-52-7	7.15967	1116.6266	248.71	-92.60	161.70	1,2
413	SiCl4	silicon tetrachloride	10026-04-7	7.00381	1205.9453	235.64	-68.85	233.85	1,2
414	SiF4	silicon tetrafluoride	7783-61-1	6.72479	459.3741	215.21	-156.00	-14.15	1,2
415	SiHBr3	silane tribromo	7789-57-3	6.97115	1382.4891	228.99	-30.50	344.35	1,2
416	SiHCl3	silane trichloro	10025-78-2	7.10586	1191.2883	250.11	-128.20	205.85	1,2
417	SiHF3	silane trifluoro	13465-71-9	6.50485	466.9478	223.85	-152.00	3.50	1,2
418	SiH2Br2	silane dibromo	13768-94-0	7.05269	1295.7656	244.60	-60.90	286.09	1,2
419	SiH2Cl2	silane dichloro	4109-96-0	6.98538	1032.4344	243.23	-122.00	175.85	1,2
420	SiH2F2	silane difluoro	13824-36-7	6.68670	609.3862	237.92	-146.70	37.60	1,2
421	SiH2I2	silane diiodo	13760-02-6	7.36622	2014.1634	299.05	3.80	410.66	1,2
422	SiH3Br	silane monobromo	13465-73-1	7.24230	1152.7870	261.91	-85.70	182.00	1,2
423	SiH3Cl	silane monochloro	13465-78-6	6.76710	797.8130	235.69	-117.80	123.64	1,2
424	SiH3F	silane monofluoro	13537-33-2	6.29474	412.9780	218.97	-153.00	12.72	1,2
425	SiH3I	silane iodo	13598-42-0	7.04589	1204.6095	243.82	-53.00	251.44	1,2
426	SiH4	silane	7803-62-5	7.12525	704.8874	278.22	-185.00	-3.45	1,2
427	SiI4	silane tetraiodo	13465-84-4	6.97269	1909.4768	201.81	120.50	256.85	1,2
428	SiO2	silicon dioxide (alpha quartz)	14808-60-7	9.87451	16418.2297	117.52	1609.85	2230.05	2
429	Si2CH3Cl5	silane pentachloromethyl-di	26980-40-5	7.91847	2186.1100	281.29	34.85	37.85	1,2
430	Si2CH7Cl	silane [(chlorosilyl)methyl]-	35483-45-5	7.83841	1674.1300	273.15	-0.15	64.54	1,2
431	Si2CH7Cl	silane (chloromethyl)-di	54713-74-5	8.09282	1875.0700	273.15	-23.15	86.61	1,2
432	Si2C2H4Cl6	1,2-bis(trichlorosilyl)ethane	2504-64-5	7.82031	2242.0000	251.00	90.85	202.89	1,2
433	Si2C2H4Cl6	1,1,1,3,3,3-hexachloro-2-methyl-1,3-disilapropane	18076-92-1	8.10890	2425.0000	273.15	54.85	190.69	1,2
434	Si2C2H6Cl4	silane 1,2,2,2-tetrachloro-1,1-dimethyldi	26980-43-8	7.14487	1680.7900	236.84	26.85	157.34	1,2
435	Si2C2H9FO	1-fluoro-1,3-dimethyldisiloxane	35192-38-2	6.32253	1162.7400	274.47	-78.15	63.37	1,2
436	Si2Cl3F3O	disiloxane, trichlorotrifluoro	---	9.03398	2311.2185	332.87	-38.00	42.74	1,2
437	Si2Cl6	disilane hexachloro	13465-77-5	7.36013	1615.6257	215.69	4.00	145.00	1,2
438	Si2Cl6O	disiloxane, hexachloro	14986-21-1	7.12363	1465.7767	210.43	-5.00	135.60	2
439	Si2F6	disilane hexafluoro	---	12.22960	2464.6767	282.54	-81.00	-18.90	1,2
440	Si2H6	disilane	1590-87-0	7.00941	1015.4056	260.09	-129.30	158.85	1,2
441	Si2H6O	disiloxane	13597-73-4	7.42748	1137.4567	265.37	-112.50	-15.20	1,2
442	Si3Cl8	octachlorotrisilane	13596-23-1	8.65954	2872.1180	285.62	46.30	211.40	1,2
443	Si3H8	trisilane	7783-26-8	7.22385	1329.1653	253.14	-68.90	52.90	1,2
444	Si3H9N	trisilazane	---	7.15693	1247.8657	243.12	-68.70	48.70	1,2
445	Si4H10	tetrasilane	7783-29-1	6.54087	1122.5646	199.31	-27.70	107.40	1,2
446	Sm	samarium	7440-19-9	6.72424	7242.8476	90.48	700.00	2000.00	1,2
447	Sn	tin (white)	7440-31-5	8.54915	16655.8123	336.40	1150.00	2800.00	1,2
448	SnBr4	stannic bromide	7789-67-5	7.14579	1846.6483	227.98	58.30	205.00	1,2
449	SnCl2	stannous chloride	7772-99-8	7.51627	3705.3738	176.36	316.00	623.00	1,2
450	SnCl4	stannic chloride	7646-78-8	6.85720	1294.1344	211.30	-22.70	114.15	1,2
451	SnH4	stannane	2406-52-2	7.15934	937.8599	271.00	-140.00	-51.80	1,2
452	SnI4	stannic iodide	7790-47-8	6.06085	1628.2894	147.69	156.00	364.35	1,2
453	SnS	tin sulfide	1314-95-0	9.97000	10460.0000	273.15	726.85	1026.85	1,2
454	Sr	strontium	7440-24-6	6.83233	6128.6851	168.97	523.00	1382.00	1,2
455	SrF2	strontium fluoride	7783-48-4	8.06789	12904.6141	3.46	1477.00	2426.85	1,2
456	SrO	strontium oxide	1314-11-0	7.51441	9499.3367	-803.52	1750.00	2700.00	1,2
457	Ta	tantalum	7440-25-7	8.73303	31263.6507	-115.81	3024.00	5458.00	1,2
458	TaBr5	tantalum bromide	13451-11-1	8.14000	3260.0000	273.15	265.00	336.85	1,2
459	TaCl5	tantalum chloride	7721-01-9	11.98340	4654.0000	273.15	188.13	238.13	1,2
460	TaF5	tantalum fluoride	7783-71-3	6.45981	1797.9105	273.15	119.00	249.20	1,2
461	Tb	terbium	7440-27-9	7.41799	14824.6184	37.37	1516.10	3230.00	1,2
462	Tc	technetium	7440-26-8	26.08539	236826.8849	5941.04	2400.00	4400.00	1,2
463	Tc2O7	technetium heptoxide	---	9.00000	3570.0000	273.15	118.85	306.85	1,2
464	Te	tellurium	13494-80-9	6.63850	4084.3774	98.94	502.00	988.00	1,2
465	TeCl2	tellurium dichloride	10025-71-5	8.51000	3350.0000	273.15	208.00	316.85	1,2
466	TeCl4	tellurium tetrachloride	10026-07-0	8.59582	3538.9650	232.24	233.00	387.00	1,2
467	TeF4	tellurium tetrafluoride	15192-26-4	5.64000	1787.0000	273.15	129.00	196.85	1,2
468	TeF6	tellurium hexafluoride	7783-80-4	8.65918	1258.5289	256.70	-111.30	-38.90	1,2
469	TeO2	tellurium dioxide	7446-07-3	12.76126	15000.0000	273.15	1215.00	1245.00	1,2
470	Te2F10	tellurium decafluoride	53214-07-6	6.90180	1150.0000	227.00	-33.15	89.85	1,2
471	Th	thorium	7440-29-1	8.08673	25785.3686	165.09	2360.00	4788.00	1,2
472	ThBr4	thorium bromide	13453-49-1	9.56000	7550.0000	273.15	679.00	852.85	1,2
473	ThCl4	thorium chloride	10026-08-1	9.57000	7980.0000	273.15	770.00	912.85	1,2
474	ThI4	thorium iodide	7790-49-0	9.09000	6890.0000	273.15	582.85	833.85	1,2
475	ThO2	thorium oxide	1314-20-1	9.64285	31600.0000	273.15	4350.00	4400.00	1,2
476	Ti	titanium	7440-32-6	8.90223	20948.9918	190.76	1709.00	3285.00	1,2
477	TiBr4	titanium bromide	7789-68-6	6.97018	1791.8855	205.54	39.00	228.85	1,2
478	TiCl2	titanium chloride	10049-06-6	8.35740	10003.3561	174.30	1035.00	1426.85	1,2
479	TiCl4	titanium tetrachloride	7550-45-0	7.29500	1668.8244	242.21	-24.10	364.85	1,2
480	TiF4	titanium fluoride	7783-63-3	10.87761	4455.4168	273.15	234.00	284.00	1,2

Table 2 Vapor Pressure – Inorganic Compounds

$\log_{10} P = A - B/(T + C)$ (P - mmHg, T - C)

NO	FORMULA	NAME	CAS No	A	B	C	TMIN	TMAX	code
481	TiI4	titanium iodide	7720-83-4	7.57600	3054.0000	273.15	156.85	369.85	1,2
482	Tl	thallium	7440-28-0	7.58000	8060.0496	229.50	609.00	1485.00	1,2
483	TlBr	thallous bromide	7789-40-4	7.32435	4432.7693	178.58	490.00	819.00	1,2
484	TlCl	thallium chloride	7791-12-0	-5.69389	6065.8547	-1427.41	504.00	720.00	1,2
485	TlF	thallium fluoride	7789-27-7	9.88714	6902.2858	273.15	662.00	712.00	1,2
486	TlI	thallium iodide	7790-30-9	7.34352	4374.7071	156.28	440.00	824.00	1,2
487	Tm	thulium	7440-30-4	7.82243	10435.3130	205.05	844.00	1297.00	1,2
488	U	uranium	7440-61-1	7.77000	23272.8375	600.00	2052.00	4129.00	1,2
489	UBr3	uranium bromide	13470-19-4	9.06095	10506.7518	178.05	727.00	926.85	1,2
490	UBr4	uranium bromide	13470-20-7	8.12161	4834.9448	143.94	519.00	766.85	1,2
491	UCl4	uranium chloride	10026-10-5	9.01541	5767.4399	147.57	590.00	770.00	1,2
492	UCl6	uranium chloride	13763-23-0	7.87988	4000.0000	273.15	477.00	527.00	1,2
493	UF4	uranium fluoride	10049-14-6	8.93038	9333.8078	95.95	1036.00	1400.00	1,2
494	UF6	uranium fluoride	7783-81-5	8.47645	1568.8742	223.87	-38.80	56.50	1,2
495	UI4	uranium iodide	13470-22-9	8.81552	5446.7685	159.45	506.00	726.85	1,2
496	V	vanadium	7440-62-2	9.18000	22303.0522	133.69	1828.00	3406.85	1,2
497	VCl4	vanadium tetrachloride	7632-51-1	7.32955	1823.3734	258.01	-25.70	423.85	1,2
498	VF5	vanadium fluoride	7783-72-4	10.43000	2423.0000	273.15	19.50	45.00	1,2
499	VOCl3	vanadium oxytrichloride	7727-18-6	7.02483	1518.9432	239.69	-79.50	126.85	1,2
500	V2O5	vanadium oxide	1314-62-1	5.05000	7100.0000	273.15	670.00	1226.85	1,2
501	W	tungsten	7440-33-7	9.98307	40387.9650	131.64	3204.00	5555.00	1,2
502	WCl4	tungsten chloride	13470-13-8	8.19500	3253.0000	273.15	281.85	326.85	1,2
503	WCl5	tungsten chloride	13470-14-9	7.72000	2760.0000	273.15	242.00	280.00	1,2
504	WCl6	tungsten chloride	13283-01-7	7.87000	3050.0000	273.15	275.00	340.00	1,2
505	WF6	tungsten fluoride	7783-82-6	7.07104	907.7240	199.53	-71.60	17.10	1,2
506	WOF4	tungsten oxytetrafluoride	13520-79-1	9.32296	2709.3726	234.67	2.00	185.90	1,2
507	WO3	tungsten oxide	1314-35-8	14.53875	24600.0000	273.15	1787.00	1837.00	1,2
508	Xe	xenon	7440-63-3	7.02795	701.1036	277.06	-111.75	16.55	1,2
509	XeF2	xenon difluoride	13709-36-9	6.96427	1073.3758	148.51	2.90	357.85	1,2
510	XeF4	xenon tetrafluoride	13709-61-0	10.91390	3095.0600	269.56	35.85	117.85	1,2
511	Y	yttrium	7440-65-5	8.00530	17991.8060	165.95	1610.10	3345.00	1,2
512	Yb	ytterbium	7440-64-4	7.28084	6060.1722	181.30	463.00	1196.00	1,2
513	Zn	zinc	7440-66-6	7.57409	5241.2423	203.39	337.00	912.00	1,2
514	ZnBr2	zinc bromide	7699-45-8	8.49304	4782.6046	185.14	397.85	561.85	1,2
515	ZnCl2	zinc chloride	7646-85-7	8.50869	5054.5539	166.13	428.00	732.00	1,2
516	ZnF2	zinc fluoride	7783-49-5	6.56144	4368.9367	-312.99	970.00	1500.00	1,2
517	ZnI2	zinc iodide	10139-47-6	18.01392	19725.9709	678.50	301.00	625.00	1,2
518	ZnO	zinc oxide	1314-13-2	9.97977	15781.3538	273.15	1499.95	1949.95	2
519	ZnS	zinc sulfide (wurtzite)	1314-98-3	12.46830	13980.0000	273.15	1135.00	1185.00	2
520	ZnSO4	zinc sulfate	7733-02-0	8.74032	2215.7753	273.15	20.00	105.00	2
521	Zr	zirconium	7440-67-7	9.12202	28773.5320	201.25	2366.00	4409.00	1,2
522	ZrBr4	zirconium bromide	13777-25-8	10.28269	4025.9186	183.91	207.00	360.00	1,2
523	ZrCl4	zirconium chloride	10026-11-6	11.44848	4801.8756	229.46	190.00	331.00	1,2
524	ZrF4	zirconium fluoride	7783-64-4	15.61264	15000.0000	273.15	875.00	905.00	1,2
525	ZrI4	zirconium iodide	13986-26-0	11.20709	5397.4287	217.24	264.00	431.00	1,2

Code: 1 - data, 2 - estimate, 3 - rough estimate.

P - vapor pressure, mmHg.

A, B, and C - Antoine coefficients for chemical compound.

T - temperature, °C.

TMIN - minimum temperature, °C.

TMAX - maximum temperature, °C.